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Cramér–von Mises Tests for Independence*

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In this paper recent results of Gregory [(1977) *Ann. Statist.* 5 110–123] are used to obtain the asymptotic null distribution of a weighted Cramér–von Mises type test for independence. We use approximate Bahadur slopes to find good weight functions for certain alternatives. Some percentage points of the asymptotic distribution are given.

1. INTRODUCTION

Let $(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$ be independent and identically distributed random vectors with joint distribution function (*df*) $H(x, y)$ and marginal *dfs* $F(x)$ and $G(y)$ respectively (all assumed to be continuous). We wish to test the hypothesis that X and Y are independent, i.e.,

$$H_0 : H = F \cdot G.$$

Let H_n, F_n, G_n be the empirical *dfs* of $\{(X_i, Y_i)\}$, $\{X_i\}$ and $\{Y_i\}$ respectively. Since these are consistent estimators of H, F, G, an intuitively appealing test statistic for H_0 is given by

$$T_n = n \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (H_n(x, y) - F_n(x)\, G_n(y))^2\, W_1(F_n^*(x))\, W_2(G_n^*(y)) \times dF_n(x)\, dG_n(y)$$

with W_1 and W_2 nonnegative weight functions on (0, 1) and $F_n^*(x) = nF_n(x)/(n+1)$, $G_n^*(y) = nG_n(y)/(n+1)$. We reject H_0 for large values of T_n. The statistic studied by Blum, Kiefer and Rosenblatt [2] closely corresponds to the case $W_1 = W_2 \equiv 1$.

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In terms of ranks, the statistic can be written in the following alternative form. Let R_{1i} be the rank of X_i in $X_1, X_2, \ldots, X_n$ and R_{2i} the rank of Y_i in $Y_1, Y_2, \ldots, Y_n$. It then follows easily that

$$T_n = n^{-1} \sum_{k,l=1}^{n} \left(n^{-1} \sum_{i=1}^{n} I(R_{1i} \leqslant k;\, R_{2i} \leqslant l) - \frac{k}{n} \cdot \frac{l}{n} \right)^2 \times W_1\left(\frac{k}{n+1}\right) W_2\left(\frac{l}{n+1}\right)$$

with $I(\cdot)$ denoting the indicator function. In this form T_n can easily be calculated.

Using the probability integral transformation it follows that under H_0 T_n is distribution free so we may take H the uniform df on $(0, 1) \times (0, 1)$ and F and G both uniform on $(0, 1)$. Under H_0 the test statistic becomes

$$n \int_0^1 \int_0^1 (H_n(u, v) - F_n(u)\, G_n(v))^2\, W_1(F_n^*(u))\, W_2(G_n^*(v))\, dF_n(u)\, dG_n(v)$$

where H_n, F_n, G_n are now based on $\{(U_i, V_i)\}$ with U_i, V_i independent, each uniformly distributed on $(0, 1)$.

In the next section we show how the results of Gregory [6] can be used to obtain the asymptotic distribution of T_n under H_0. In section 3 we make a number of remarks about the kernel whose eigenvalues characterize the asymptotic distribution of T_n and in section 4 we consider certain alternatives and find weight functions which are optimal in the sense of approximate Bahadur slope. Some percentage points of the asymptotic distribution are given in Section 5.

2. The Asymptotic Null Distribution of T_n

In this section we obtain the asymptotic distribution of T_n under H_0. We need the following conditions on W_1 and W_2.

(A1) W_i is continuous and differentiable on $(0, 1)$, $i = 1, 2$.

(A2) There exist constants C_i, D_i, $i = 1, 2$ such that

$$W_i(u) \leqslant C_i(u(1-u))^{-2}\, |\log u(1-u)|^{-1} \qquad i = 1, 2$$

$$|W_i'(u)| \leqslant D_i(u(1-u))^{-3}\, |\log u(1-u)|^{-1} \qquad i = 1, 2.$$

Remark. The bounds given in (A2) are primarily motivated by the weight function $\phi(\Phi^{-1}(u))^{-2}$ considered below.

Let

$$T_{1n} = n \int_0^1 \int_0^1 (H_n(u, v) - vF_n(u) - uG_n(v) + uv)^2\, W_1(u)\, W_2(v)\, du\, dv.$$

We show in the next lemma that $T_n - T_{1n} = o_p(1)$ and in the theorem we show how to apply Gregory's result to T_{1n}.

Throughout this section and the next one we assume H_0 to be true.

LEMMA. *As* $n \to \infty$, $T_n - T_{1n} = o_p(1)$.

Proof. Let $R_n = T_n - T_{1n}$, then R_n can be written as

$$R_n = R_{1n} + R_{2n} + R_{3n}$$

where

$$R_{1n} = n \int_0^1 \int_0^1 g_{1n}^2(u, v)(W_1(F_n^*(u))\, W_2(G_n^*(v)) - W_1(u)\, W_2(v))\, dF_n(u)\, dG_n(v)$$

$$R_{2n} = n \int_0^1 \int_0^1 (g_{1n}^2(u, v) - g_{2n}^2(u, v))\, W_1(u)\, W_2(v)\, dF_n(u)\, dG_n(v).$$

$$R_{3n} = n \int_0^1 \int_0^1 g_{2n}^2(u, v)\, W_1(u)\, W_2(v)(dF_n(u)\, dG_n(v) - du\, dv)$$

with

$$g_{1n}(u, v) = H_n(u, v) - F_n(u)\, G_n(v)$$
$$g_{2n}(u, v) = H_n(u, v) - vF_n(u) - uG_n(v) + uv.$$

Note that $ng_{1n}^2(u, v)$, $ng_{2n}^2(u, v)$, $n^{\frac{1}{2}}(F_n(u) - u)$ and $n^{\frac{1}{2}}(G_n(v) - v)$ all converge in distribution so the lemma is intuitively clear. Because of the heavy weight W_1 and W_2 can place near 0 and 1 (by (A_2)), R_{1n} is the most difficult term to handle. We will prove that $R_{1n} = o_p(1)$; the other terms can be handled in a similar fashion.

Let

$$V_n(i, j) = H_n(U_{(i)}, V_{(j)}) - ij/n^2$$

with $\{U_{(i)}\}$ and $\{V_{(j)}\}$ the order statistics of the U and V samples. Then

$$\begin{aligned}
R_{1n} &= n^{-1} \sum_{i,j=1}^{n} V_n^2(i, j) \left(W_1\left(\frac{i}{n+1}\right) W_2\left(\frac{j}{n+1}\right) - W_1(U_{(i)})\, W_2(V_{(j)})\right) \\
&= -n^{-1} \sum_{i,j} V_n^2(i, j)\, W_1\left(\frac{i}{n+1}\right)\left(W_2(V_{(j)}) - W_2\left(\frac{j}{n+1}\right)\right) \\
&\quad - n^{-1} \sum_{i,j} V_n^2(i, j)\, W_2\left(\frac{j}{n+1}\right)\left(W_1(U_{(i)}) - W_1\left(\frac{i}{n+1}\right)\right) \\
&\quad - n^{-1} \sum_{i,j} V_n^2(i, j) \left(W_1(U_{(i)}) - W_1\left(\frac{i}{n+1}\right)\right)\left(W_2(V_{(j)}) - W_2\left(\frac{j}{n+1}\right)\right) \\
&= -(R_{11n} + R_{12n} + R_{13n}) \quad \text{(say)}.
\end{aligned}$$

We will only consider R_{11n}; the other two terms can be treated analogously.

Let $0 < \alpha < 1$; by splitting the summation over i into three parts, viz., $1 \leqslant i \leqslant [n\alpha]$, $[n\alpha] + 1 \leqslant i \leqslant [n(1-\alpha)]$, $[n(1-\alpha)] + 1 \leqslant i \leqslant n$ and likewise for j, we split R_{11n} into nine parts of which the most difficult to handle are the ones where $1 \leqslant i, j \leqslant [n\alpha]$ and $[n(1-\alpha)] + 1 \leqslant i, j \leqslant n$. We will only consider the first one; the second one follows by symmetry.

Let $\alpha_n = [n\alpha]$ and

$$R_{11n}(\alpha) = n^{-1} \sum_{i,j=1}^{\alpha_n} V_n^2(i,j)\, W_1\left(\frac{i}{n+1}\right)\left(W_2(V_{(j)}) - W_2\left(\frac{j}{n+1}\right)\right).$$

Applying a Taylor expansion to first order, assumption (A2) and proposition 4 of Chernoff, Gastwirth and Johns [3], we find that with probability at least $1 - \epsilon$ for α small enough

$$|R_{11n}(\alpha)| \leqslant n^{-1} \sum_{i,j \leqslant \alpha_n} V_n^2(i,j) W_1\left(\frac{i}{n+1}\right)\left| V_{(j)} - \frac{j}{n+1}\right| u_{jn}(\epsilon)^{-3} |\log u_{jn}(\epsilon)|^{-1}$$

with $\{u_{jn}(\epsilon)\}$ the numbers in Chernoff, Gastwirth and Johns [3]. Now, by straightforward but lengthy calculations it follows that there exist constants A_1 and A_2 such that for $1 \leqslant i, j \leqslant n$,

$$EV_n^2(i,j) \leqslant A_1 n^{-1} \frac{i}{n}\left(1 - \frac{i}{n}\right)\frac{j}{n}\left(1 - \frac{j}{n}\right)$$

and

$$EV_n^4(i,j) \leqslant A_2 n^{-2} \left(\frac{i}{n}\left(1 - \frac{i}{n}\right)\frac{j}{n}\left(1 - \frac{j}{n}\right)\right)^2.$$

Using this, the properties of $\{u_{jn}(\epsilon)\}$ and Schwarz's inequality, we have with probability at least $1 - \epsilon$, for $0 < \delta < 1$, that

$$\begin{aligned}
&E\,|R_{11n}(\alpha)| \\
&\leqslant K_1 n^{-1} \sum_{i,j \leqslant \alpha_n} n^{-1} \frac{ij}{n^2}\left(\frac{n}{i}\right)^2 \left|\log \frac{i}{n}\right|^{-1} n^{-1/2}\left(\frac{j}{n}\right)^{1/2}\left(\frac{n}{j}\right)^3 \left|\log \frac{j}{n}\right|^{-1} \\
&\leqslant K_1 \sum_{i \leqslant \alpha_n} i^{-1} \left|\log \frac{i}{n}\right|^{-1} \sum_{j \leqslant \alpha_n} j^{-3/2} \left|\log \frac{j}{n}\right|^{-1} \\
&\leqslant K_2\, |\log \alpha|^{-1}\, O(\log n)\left(\sum_{j \leqslant [n^\delta]} j^{-3/2}\left|\log \frac{j}{n}\right|^{-1} + \sum_{[n^\delta] < j \leqslant \alpha_n} j^{-3/2}\left|\log \frac{j}{n}\right|^{-1}\right) \\
&\leqslant K_3\, |\log \alpha|^{-1}\, O(\log n)(O((\log n)^{-1}) + O(n^{-\delta/2}))
\end{aligned}$$

with K_1, K_2, K_3 constants. Thus with probability at least $1 - \epsilon$ we have

$$\lim_{\alpha \to 0} \overline{\lim_{n \to \infty}}\, E\,|R_{11n}(\alpha)| = 0$$

and a similar result for the other terms in R_{11n}; so $R_{11n} = o_p(1)$. The other terms can be treated likewise and the lemma follows. ∎

Now, let $\mathbf{s} = (s_1, s_2)$, $\mathbf{t} = (t_1, t_2)$, $\mathbf{z} = (z_1, z_2)$, and

$$Q(\mathbf{s}, \mathbf{t}) = \int_0^1 \int_0^1 q(\mathbf{s}, \mathbf{z})\, q(\mathbf{t}, \mathbf{z})\, dz_1\, dz_2$$

with

$$q(\mathbf{s}, \mathbf{t}) = (I(s_1 \leqslant t_1) - t_1)(I(s_2 \leqslant t_2) - t_2)(W_1(t_1)\, W_2(t_2))^{\frac{1}{2}}.$$

We then have the following

THEOREM. *Under conditions* (A1) *and* (A2) *there exist constants* $\{\mu_n\}$ *such that*

$$T_n - \mu_n \xrightarrow{D} \sum_{j,k=1}^{\infty} \gamma_{jk}(N_{jk}^2 - 1)$$

where $\{N_{jk}\}$ *are i.i.d.* $N(0, 1)$ *and* $\{\gamma_{jk}\}$ *are the eigenvalues of* Q.

Proof. Using the definitions of H_n, F_n, G_n, we can write

$$\begin{aligned} T_{1n} &= n \int_0^1 \int_0^1 \left[n^{-1} \sum_{j=1}^{n} (I(U_j \leqslant u;\, V_j \leqslant v) - vI(U_j \leqslant u) - uI(V_j \leqslant v)) + uv \right]^2 \\ &\quad \times W_1(u)\, W_2(v)\, du\, dv \\ &= n \int_0^1 \int_0^1 \left[n^{-1} \sum_j (I(U_j \leqslant u) - u)(I(V_j \leqslant v) - v) \right]^2 W_1(u)\, W_2(v)\, du\, dv. \end{aligned}$$

Squaring, changing the order of summation and integration and writing $\mathbf{Z}_i = (U_i, V_i)$ we have

$$T_{1n} = n^{-1} \sum_{i,j=1}^{n} Q(\mathbf{Z}_i, \mathbf{Z}_j).$$

Note that by using (A2) Q is square integrable; it is also symmetric in its arguments and

$$\int_0^1 \int_0^1 Q(\mathbf{s}, \mathbf{t})\, ds_1\, ds_2 = 0.$$

To apply Theorem 2.1 of Gregory [6] we have to show that (see [6, (3.1)])

$$\lim_{t \to \infty} tP(Q(\mathbf{Z}_1, \mathbf{Z}_1) > t) = 0.$$

First note that by rearranging terms, we have

$$Q(\mathbf{s}, \mathbf{t}) = C_{W_1}(s_1, t_1)\, C_{W_2}(s_2, t_2)$$

with

$$C_W(s, t) = \int_0^1 (I(s \leqslant u) - u)(I(t \leqslant u) - u)\, W(u)\, du.$$

By using (A2) we have, for $u < \alpha < 1$, $i = 1, 2$,

$$\begin{aligned} C_{W_i}(u, u) &= \int_0^1 (I(u \leqslant z) - z)^2\, W_i(z)\, dz \\ &= \int_0^u z^2 W_i(z)\, dz + \int_u^\alpha (1 - z)^2\, W_i(z)\, dz + \int_\alpha^1 (1 - z)^2\, W_i(z)\, dz \\ &\leqslant b + a_1 \int_u^\alpha z^{-2} \mid \log z \mid^{-1} dz \end{aligned}$$

for constants a_1, b. Using l'Hospital, we have

$$\int_u^\alpha z^{-2} \mid \log z \mid^{-1} dz \sim u^{-1} \mid \log u \mid^{-1} \qquad \text{as} \quad u \to 0.$$

Thus, for u small enough, we have

$$C_{W_i}(u, u) \leqslant b + au^{-1} \mid \log u \mid^{-1}, \qquad i = 1, 2. \tag{1}$$

An analogous result holds for u close to 1. Let $\mathscr{L}$ denote Lebesgue measure on the unit square. We wish to consider

$$\mathscr{L}((u, v) : C_{W_1}(u, u)\, C_{W_2}(v, v) > t;\, 0 < u, v < 1).$$

Split the unit square into nine parts, viz., $0 < u < \alpha$, $\alpha < u < 1 - \alpha$, $1 - \alpha < u < 1$ and likewise for v. We will only consider the case where $0 < u, v < \alpha$; the case where $1 - \alpha < u, v < 1$ follows by symmetry while the other 7 parts can be treated in a similar (albeit easier) way. Take α so small that (1) holds for $u \leqslant \alpha$. We then want to consider $\mathscr{L}(A(t))$ where

$$A(t) = ((u, v): (b + au^{-1} \mid \log u \mid^{-1})(b + av^{-1} \mid \log v \mid^{-1}) > t;\, 0 < u, v < \alpha).$$

By splitting $A(t)$ into four sets, viz., $0 < u < t^{-\frac{1}{2}} \mid \log t \mid^{-\frac{1}{2}}$, $t^{-\frac{1}{2}} \mid \log t \mid^{-\frac{1}{2}} < u < \alpha$ and likewise for v, it follows fairly easily that $\mathscr{L}(A(t)) = o(t^{-1})$ as $t \to \infty$, and the same result holds for the other terms. The theorem follows from this and the lemma by applying Gregory's result.

Remarks. 1. The reason for using double subscripts on $\{\gamma_{jk}\}$, $\{N_{jk}\}$ will be clear from the next section.

2. Some known weight functions covered by the theorem are those associated with the classical orthogonal polynomials; see, e.g., De Wet and Venter [4, p. 385].

3. Note that the results of Blum, Kiefer and Rosenblatt [2] do not cover the case where $\sum_{j,k}\gamma_{jk} = \infty$, while our result only requires $\sum_{j,k}\gamma_{jk}^2 < \infty$.

4. In section 5 we show how to invert the characteristic function of the random variable $\sum_{j,k}\gamma_{jk}(N_{jk}^2 - 1)$ to obtain approximate percentage points for the distribution of T_n.

5. The constants $\{\mu_n\}$ are given by

$$\mu_n = \int_0^1 \int_0^1 Q(\mathbf{t}, \mathbf{t})\, I(-n < Q(\mathbf{t}, \mathbf{t}) < n)\, dt_1\, dt_2$$

and can be calculated by numerical integration for given W_1 and W_2.

3. The Kernel Q

We now make some remarks concerning the kernel Q.

3.1. As noted above, we have

$$Q(\mathbf{s}, \mathbf{t}) = C_{W_1}(s_1, t_1)\, C_{W_2}(s_2, t_2)$$

with

$$C_W(s, t) = \int_0^1 (I(s \leqslant u) - u)(I(t \leqslant u) - u)\, W(u)\, du.$$

The kernel $C_W(s, t)$ is exactly the kernel obtained in the modified Cramer-von Mises goodness of fit statistic, see, e.g., Gregory [6, p. 114]. Now, let $(\gamma_k^{(i)}, g_k^{(i)})$, $k = 1, 2,\ldots$ be an eigenpair of C_{W_i}, $i = 1, 2$ and put

$$\phi_{jk}(\mathbf{t}) = g_j^{(1)}(t_1)\, g_k^{(2)}(t_2),$$

and

$$\gamma_{jk} = \gamma_j^{(1)}\gamma_k^{(2)}.$$

It then follows that (γ_{jk}, ϕ_{jk}), $j, k = 1, 2,\ldots$ is an eigenpair of Q. Thus, to solve the integral equation to obtain γ_{jk} we need only solve the integral equations for $\gamma_j^{(1)}$ and $\gamma_k^{(2)}$; these have been done for quite a number of weight functions (see De Wet and Venter [4]).

3.2. First note that if W_1 and W_2 are such that $\sum_{j,k}\gamma_{jk} < \infty$ then $\mu_n \to \sum_{j,k}\gamma_{jk}$, which means that

$$T_n \xrightarrow{D} \sum_{j,k=1}^{\infty} \gamma_{jk} N_{jk}^2.$$

For the present consider this to be the case. The so-called stochastic process approach as used by, e.g., Blum *et al.* [2] then gives

$$T_n \xrightarrow{D} \int_0^1 \int_0^1 B(u, v)^2 \, du \, dv$$

where B is a zero mean Gaussian process with covariance kernel

$$\begin{aligned} K(\mathbf{s}, \mathbf{t}) &= EB(s_1, s_2) B(t_1, t_2) \\ &= (\min(s_1, t_1) - s_1 t_1)(W_1(s_1) W_1(t_1))^{\frac{1}{2}}(\min(s_2, t_2) - s_2 t_2)(W_2(s_2) W_2(t_2))^{\frac{1}{2}} \end{aligned}$$

Let $\{\lambda_{jk}\}$ and $\{h_{jk}\}$ be the eigenvalues and eigenfunctions of K, then $B(u, v)$ can be represented by

$$B(u, v) = \sum_{j,k=1}^{\infty} \lambda_{jk}^{\frac{1}{2}} h_{jk}(u, v) N_{jk} .$$

Substituting it follows that

$$\int_0^1 \int_0^1 B(u, v)^2 \, du \, dv \overset{D}{=} \sum_{j,k=1}^{\infty} \lambda_{jk} N_{jk}^2 .$$

It follows that

$$\sum_{j,k} \gamma_{jk} N_{jk}^2 \overset{D}{=} \sum_{j,k} \lambda_{jk} N_{jk}^2 \tag{2}$$

with $\{\gamma_{jk}\}$ the eigenvalues of Q and $\{\lambda_{jk}\}$ the eigenvalues of K. We now indicate the relationship that exists between Q and K. Let T_Q be the integral operator corresponding to Q, i.e.,

$$(T_Q g)(\mathbf{s}) = \int_0^1 \int_0^1 Q(\mathbf{s}, \mathbf{t}) \, g(\mathbf{t}) \, dt_1 \, dt_2$$

and likewise for T_q.

Since

$$Q(\mathbf{s}, \mathbf{t}) = \int_0^1 \int_0^1 q(\mathbf{s}, \mathbf{z}) \, q(\mathbf{t}, \mathbf{z}) \, dz_1 \, dz_2$$

it follows that

$$T_Q = T_q T_q^* \tag{3}$$

with T_q^* denoting the adjoint of T_q. From the definition of K it follows directly that

$$T_K = T_q^* T_q \tag{4}$$

which shows how Q and K are related (see also De Wet and Venter [5] and Neuhaus [10]). From this it follows immediately that Q and K have the same eigenvalues, as one would suspect from (2). Let g and h be the eigenfunctions of Q and K corresponding to an eigenvalue γ. It then follows directly from (3) and (4) that

$$h = T_q^* g / \| T_q^* g \|, \qquad \text{where} \quad \| \cdot \| \text{ denotes norm.}$$

4. Optimal Weight Functions

The efficiency of the class of test statistics T_n depends of course on the choice of W_1 and W_2. In Gregory [7] the approximate Bahadur efficiency of a class of quadratic test statistics, which includes our statistics, is discussed. He obtains sufficient conditions for a test to be optimal in the class of all quadratic tests. We apply these results below to obtain optimal weight functions in two particular cases.

Consider a sequence of alternative hypotheses K_θ, indexed by a parameter θ, under which (X, Y) has probability distribution P_θ. For simplicity we assume H_0 to be K_0, i.e. when $\theta = 0$; thus P_0 denotes the probability distribution under H_0.

Gregory [7] considers the case of contiguous alternatives i.e.

$$dP_\theta / dP_0 = 1 + \theta h_\theta$$

where $\{h_\theta\}$ converges to h_0 as $\theta \to 0$.

Gregory's result (3.12) then says that with respect to limiting approximate Bahadur efficiency an optimal quadratic test is one which has an eigenfunction proportional to h_0 and corresponding to the largest eigenvalue.

We now apply this result to two examples. In the notation of Section 3.1 let $\phi_{11}(\mathbf{t}) = g_1^{(1)}(t_1)\, g_2^{(2)}(t_2)$ be the eigenfunction corresponding to the largest eigenvalue.

1. *Bivariate Normal Alternatives*

Let P_θ denote the bivariate normal distribution with means zero, variances one and correlation coefficient θ (we take the means zero and variances one without loss of generality).

Straightforward algebra gives

$$h_0(t_1, t_2) = -t_1 t_2 .$$

Now let

$$W_1 = W_2 = \phi(\Phi^{-1})^{-2} \tag{5}$$

with ϕ and Φ the $N(0, 1)$ density and *df*. From De Wet and Venter [4] it follows that in this case $\phi_{11}(\mathbf{t})$ is given by the product of the first Hermitian polynomials evaluated at t_1 and t_2. This is clearly proportional to the above h_0 and we conclude that (5) is the optimal choice in this case.

2. *Bivariate Exponential Alternatives*

Let P_θ be Gumbel's bivariate exponential distribution (see e.g. Johnson and Kotz [8]) with *df*

$$1 - e^{-x} - e^{-y} + e^{-x-y+\theta xy}$$

(We take the scale parameters as one without loss of generality). It follows easily that in this case

$$h_0(t_1, t_2) = (1 - t_1)(1 - t_2)$$

Take

$$W_1(u) = W_2(u) = -(1 - u)^{-2}(\log(1 - u))^{-1} \tag{6}$$

(See De Wet and Venter [4], eq. (35) for the origin of this choice). It follows again from De Wet and Venter [4] that in this case $\phi_{11}(\mathbf{t})$ is given by the product of the first Laguerre polynomials evaluated at t_1 and t_2. Since this is proportional to the above h_0, we conclude that the choice (6) is optimal in this case.

Remarks. 1. Consider the case of bivariate normal alternatives. Let $\mathbf{W}_1$ be the choice $(\phi(\Phi^{-1}(u))^{-2}, \phi(\Phi^{-1}(v))^{-2})$, $\mathbf{W}_2$ the choice $(1, 1)$ and $\mathbf{W}_3$ the choice $((u(1 - u))^{-1}, v(1 - v))^{-1})$ (the Anderson-Darling weight functions). Let e_{ij} be the limiting relative Bahadur efficiency of the statistic T_n based on $\mathbf{W}_j$ compared to the one base on $\mathbf{W}_i$. It then follows easily that

$$e_{12} = 0.826$$
$$e_{13} = 0.924.$$

Note also that the approximate slope of $r((n - 2)/(1 - r^2))^{\frac{1}{2}}$, with r the sample correlation coefficient, is $\theta^2/(1 - \theta^2)$ and for the locally most powerful rank test it is θ^2 (see Abrahamson [1]). The statistic T_n based on the optimal choice of $\mathbf{W}$ thus has a limiting efficiency of 1 relative to these two statistics.

2. The bivariate exponential distribution was also considered by Kowalski and Tarter [9] as an alternative to independence. Let $\mathbf{W}_4$ be the choice $(-(1 - u)^{-2}(\log(1 - u))^{-1}, -(1 - v)^{-2}(\log(1 - v))^{-1})$ and $\mathbf{W}_2$, $\mathbf{W}_3$ as in the previous remark. It then follows that

$$e_{42} = 0.534$$
$$e_{43} = 0.653$$

so the optimal choice gives us quite an improvement over $\mathbf{W}_2$ and $\mathbf{W}_3$ in this case.

5. Percentage Points

For both the choices (5) and (6) of $\mathbf{W}$ we have $\gamma_{jk} = (jk)^{-1}$ (see De Wet and Venter [4]) giving the limiting random variable of $T_n - \mu_n$ as

$$T = \sum_{j,k=1}^{\infty} (jk)^{-1}(N_{jk}^2 - 1).$$

The characteristic function of T can be written as

$$\prod_{j,k=1}^{\infty} (1 - 2it/jk)^{-1/2} e^{-it/jk} = r(t)\, e^{i\theta(t)}$$

with

$$r(t) = \prod_{k=1}^{\infty} (\sinh(2\pi t/k)/(2\pi t/k))^{-1/4}$$

and

$$\theta(t) = \frac{1}{2} \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} ((\tan^{-1} 2t/jk) - (2t/jk)).$$

Inversion of this characteristic function has been done numerically and a few percentage points are given in Table I. Other choices of $\mathbf{W}$ can be treated similarly. Percentage points for the case where $\mathbf{W} = (W_1, W_2)$ with $W_1 = W_2 \equiv 1$ can be found in Blum *et al.* [2].

We also need the values of μ_n to apply the test. These can be calculated fairly easily for a given $\mathbf{W}$; in Table II we give values of μ_n for the choices (5) and (6).

Remarks. 1. By using Gregory's result, the theorem can also be proved for a sequence of contiguous alternatives; this can then be used to make

TABLE I

Upper Percentage Points of T

Percentage	Percentage point
99	6.90
95	4.18
90	2.98

TABLE II

Values of the Constant μ_n

n	Choice(5)	Choice(6)
5	1.327	1.229
10	2.169	1.855
20	2.910	2.487
30	3.350	2.773
40	3.658	2.928
50	3.835	3.106
70	4.069	3.202
100	4.069	3.306

asymptotic power comparisons. We hope to report at a later stage on this as well as on the estimated small sample distribution of T_n. Note also the results of Neuhaus [10] on the asymptotic power of a fixed test at varying alternatives.

2. In this paper we only considered bivariate distributions. We will consider higher dimensions at a later stage.

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